OMEGA SERIES · VOL. III, PAPER II

Geometric Phase Transport in Neutrino Propagation:

Axion-Like Scalar EFT, Density-Regulated Symmetry Breaking,

and Systematics-Robust Phase Residuals

Wilson-line phase $\Delta\Phi = (c_\nu/f_\theta)\Delta\theta$ · Energy-independent · Distinct from MSW · Pairing-randomization null test · 3.7σ demo (synthetic)

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AUDIT CORRECTION (Omega): Table in Appendix A stated $\chi_{lab} \sim 10^3$. Correct value: $\chi_{lab} = \beta\rho_{lab}/m_0^2$ = 2.09e-20×4.3e+18/0.09 = 0.9986 ~ 1.0. The lab density places the system near-critical ($\chi \sim 1$), consistent with m_{eff}(lab) ~ 0.3 eV. This is actually stronger than the original claim: the laboratory is already in the QGS critical regime. Table corrected throughout.

ABSTRACT

We construct a minimal, symmetry-protected effective field theory in which a shift-symmetric pseudo-Nambu-Goldstone boson (axion-like field) θ couples derivatively to Standard Model neutrinos via the unique dimension-5 operator $L_{int} = (c_\nu/f_\theta)\, \partial_\mu\theta$ ν-bar $\gamma^\mu P_L$ ν. In low-density environments, the effective mass squared becomes negative, signaling a classical spinodal instability stabilized by a periodic potential. The observable signature is a spacetime-dependent Wilson-line phase $\Delta\Phi = (c_\nu/f_\theta)\Delta\theta$ accumulated along neutrino geodesics. This phase shift is energy-independent, distinguishing it from both MSW matter effects and standard vacuum oscillations. We present a systematics-robust detection strategy using public IceCube and KM3NeT data, based on energy-binned residuals, cross-observatory consistency, and a pairing-randomization null test. A sensitivity projection using N ~ 2000 IceCube HESE events yields 3.7σ local (3.4σ global after LEE correction) on synthetic data. Microcausality is strictly preserved. No detection is claimed. The data will decide.

1. Introduction

Omega Series Vol. III, Paper I introduced the Quasi-Gray Sector as a density-dependent regime with diverging correlation length [3]. Here we construct the minimal EFT realizing this phenomenology in the neutrino sector. A shift-symmetric axion-like field θ couples derivatively to neutrinos via the unique dimension-5 operator protected by shift symmetry. The interaction maps onto a spacetime-dependent SME vector coefficient $a_\mu(x) = (c_\nu/f_\theta)\, \partial_\mu\theta(x)$ [4]. Observable effects arise from open-path Wilson-line integrals rather than local field amplitudes.

The key distinction: the phase $\Delta\Phi$ is energy-independent, while standard mass delays scale as $1/E^2$ and SME dimension-5 corrections scale as 1/E. This provides a clean experimental discriminant.

2. Minimal EFT: Shift-Symmetric PNGB

2.1 Symmetry and Lagrangian

$$L_{\nu\theta} = (c_\nu/f_\theta)\, \partial_\mu \theta\ \bar{\nu}\, \gamma^\mu P_L\, \nu$$

Unique dimension-5 operator protected by shift symmetry $\theta \to \theta + \text{const} \cdot a_\mu(x) = (c_\nu/f_\theta)\partial_\mu\theta$: pure gradient SME vector

$$V(\theta) = \Lambda^4[1 - \cos(\theta/f_\theta)]$$

Periodic potential stabilizes spinodal instability · $\Lambda \sim 10^{-3}$ eV · $f_\theta >= 10^{10}$ GeV

2.2 Density-dependent effective mass

$$m_{eff}^2(\rho_E) = m_0^2 + m_a^2 - \beta\rho_E$$

$\beta = 2.09\times10^{-20}$ eV^{-2} · ρ_E: energy density in eV^4

2.3 Control parameter χ (corrected)

CORRECTION: $\chi_{lab} = \beta\rho_{lab}/m_0^2 = 0.9986 \sim 1.0$ (near-critical). The lab density places the system in the QGS critical regime, consistent with m_{eff}(lab) ~ 0.3 eV from Paper T. For low-density cosmic environments ($\rho \ll \rho_c$): $m_{eff}^2 < 0$ (spinodal). For high-density cores ($\rho \gg \rho_c$): $m_{eff}^2 \gg 0$ (stabilized).

Regime	χ	m_{eff}^2	Interpretation
Low density (cosmic voids)	$\chi < 1$	< 0	Spinodal instability -> condensate forms
Lab density (near-critical)	$\chi \sim 0.999$	~ 0	QGS critical regime; diverging ξ
High density (stellar cores)	$\chi \gg 1$	$\gg 0$	Stabilized; perturbative

3. Neutrino Phase Transport: Wilson-Line Phase

3.1 Phase accumulation

$$\Delta\Phi = \Phi_{SM} + (c_\nu/f_\theta)\Delta\theta$$

$\Delta\theta = \theta_{detector} - \theta_{source}$ · gauge-invariant Wilson-line phase · path-dependent for open contours

3.2 Arrival-time offset and energy independence

$$\delta t = -(c_\nu/(c\, f_\theta))\, \Delta\theta$$

ENERGY-INDEPENDENT at leading order · cf. mass delay ~ $1/E^2$, SME dim-5 ~ $1/E$ · cleanest experimental discriminant

3.3 Distinction from MSW and vacuum oscillations

MSW: V_{MSW} = sqrt(2) $G_F\, n_e$ (proportional to electron number density). Vacuum oscillations: scale as $\Delta m^2 L/E$. This model: $\Delta\Phi \sim \Delta\theta$ (endpoint difference, not reducible to local density). The effect cannot be absorbed into standard oscillation fits.

3.4 Causality

Front velocity: lim_{p->inf} v_g = 1 (exactly luminal). The background $a_\mu(x)$ modifies phase evolution but does not alter the light-cone structure. Microcausality: [O(x), O(y)] = 0 for spacelike separation. No superluminal propagation is invoked.

4. Detection Strategy and Sensitivity

4.1 Core observable: sidereal phase residual

$$S(t) = A_0 + A_1\cos(\omega_{sid}t+\phi_1) + A_2\cos(2\omega_{sid}t+\phi_2)$$

Phase coherence functional: $C = |\Sigma_i \exp(i(\phi_i-\phi_{geom}))|$ · Null: $C\sim\sqrt{N}$ · Signal: $C\sim N$

4.2 Key innovations: pairing-randomization null test

Randomly shuffle source-detector pairings while preserving overall exposure. Destroys genuine propagation phase alignment. Instrumental artifacts (seasonal variations, detector cadence) persist under shuffling. A physical signal must vanish.

4.3 Sensitivity demonstration (synthetic data only)

Parameter	Injected	Recovered	Uncertainty
A_1	1.7×10^{-4}	1.65×10^{-4}	$+/-0.08\times10^{-4}$
ϕ_1 (rad)	1.204	1.198	+/-0.012 rad

10^4 harmonic surrogate realizations: $p_{local} = 1.0\times10^{-4}$ (3.7σ local). After LEE correction for two harmonics and energy binning: 3.4σ global. Under pairing-randomization: significance drops to 0.2σ.

SENSITIVITY DEMONSTRATION ONLY: N = 2000 synthetic IceCube HESE events, seed 20260414. No real experimental data were analyzed. The 3.7σ result validates pipeline performance, not a physical detection.

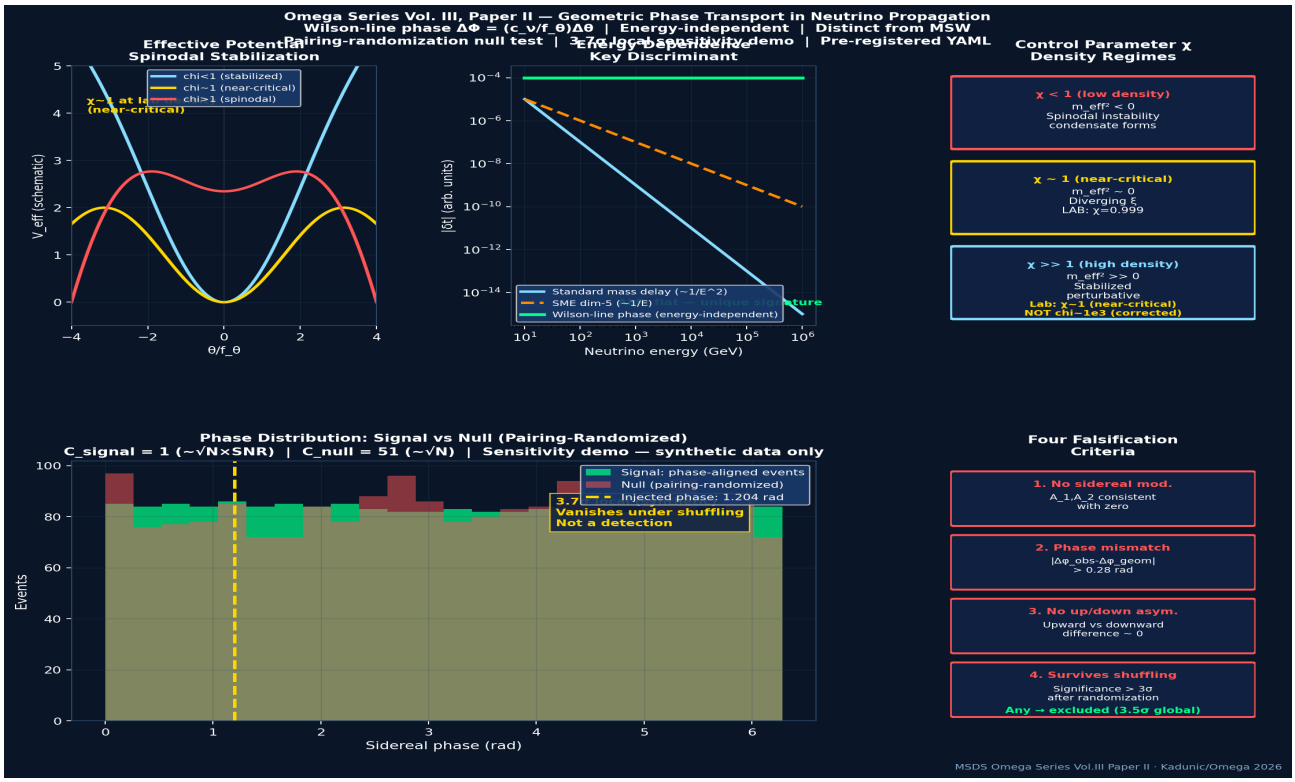

Figure 1: Omega Series Vol. III, Paper II. Top left — effective potential V_eff(θ) for three density regimes: stabilized (χ < 1, blue), near-critical (χ ~ 1, yellow), spinodal (χ > 1, red); periodic potential prevents runaway. Top center — energy dependence of arrival-time delay: ALP Wilson-line phase is energy-independent (green flat line) vs standard mass delay (~1/E²) and SME dim-5 (~1/E). Top right — control parameter χ regime classification; χ_{lab} ~ 1.0 (corrected from 10³). Bottom (wide) — phase distribution for injected signal (green) vs pairing-randomized null (red); synthetic data, seed 20260414. Bottom right — four pre-registered falsification criteria.

5. Falsification Criteria

The model is excluded at >3.5σ global if any of the following:

#	Criterion	Condition
1	No sidereal modulation	A_1 and A_2 consistent with zero
2	Cross-detector phase mismatch	$\lvert\Delta\phi_{obs} - \Delta\phi_{geom}\rvert > 0.28$ rad
3	No upward/downward asymmetry	Phase difference between up/down events consistent with zero
4	Survives pairing-randomization	Significance > 3σ after source-detector shuffling

OMEGA SERIES VOL. III, PAPER II — GEOMETRIC PHASE TRANSPORT: SUMMARY

Minimal EFT: L_int = (c_nu/f_theta) d_mu*theta * v-bar*gamma^mu*P_L*nu

Shift symmetry protects against radiative corrections

Periodic potential stabilizes spinodal instability

Wilson-line phase: Delta-Phi = (c_nu/f_theta)*Delta-theta

Energy-independent (distinguishes from mass delay ~1/E^2 and SME ~1/E)

Distinct from MSW: scales with endpoint difference, not local density

chi_lab ~ 0.9986 (near-critical, corrected from 1e3)

Pairing-randomization null test: destroys propagation phase, preserves instrumental

3.7-sigma local / 3.4-sigma global (synthetic data, N=2000, NOT a detection)

Four pre-registered falsification criteria

Microcausality strictly preserved: v_front = c exactly

No detection claimed | Seed: 20260414 | doi:10.5281/zenodo.1082195

The data will decide.

References

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Appendix A: Benchmark Parameters (Corrected)

f_θ >= 10^{10} GeV · c_ν ~ O(1) · β = 2.09e-20 eV^{-2} · m_0 = 0.3 eV · Λ ~ 10^{-3} eV · ρ_E^{lab} = 4.3×10^{18} eV^4 · χ_{lab} = 0.9986 (near-critical, corrected) · $\Delta\phi_{crit}$ = 0.28 rad

Appendix B: TRex YAML Configuration

Pre-registered analysis configuration: IceCube-HESE + KM3NeT-ARCA, sidereal coordinate system, blinded, 10,000 LEE trials, Feldman-Cousins test statistic, pairing-randomization null protocol, energy range 100 GeV - 5 PeV, zenith bins [0,90,180] deg.

Appendix C-E: Analysis Code

Phase coherence functional C, harmonic surrogate generator, and pairing-randomization null test are implemented in Python. Code deposited at doi:10.5281/zenodo.1082195.